

|                                                                                                         |                                                                                                                                                                                                                      |              |
|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Understanding the emerging and future trends in database, their issues, implementation and applications | <b>Unit VIII: Emerging Trends</b><br>8.1 Cloud Database<br>8.2 Serverless Databases<br>8.3 Vector Databases for AI/ML<br>8.4 Edge Databases (IoT Applications)<br>8.5 Ethical Considerations in Database Engineering | 4 <u>hrs</u> |
|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|

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**NCIT**

**Subject: Database Engineering (MS/ME)**

# Cloud Databases

- *A Cloud Database is a database that is built, deployed, and accessed through cloud computing platforms instead of being stored on a local server or personal computer.*
- *It allows users to store, manage, and retrieve data over the internet with high scalability, availability, and reliability.*

A cloud database is a database that runs on the cloud and can be accessed anytime, anywhere.

## Traditional Database

- Stored on **local** servers
- Limited scalability
- High hardware & maintenance cost
- Manual backup and recovery

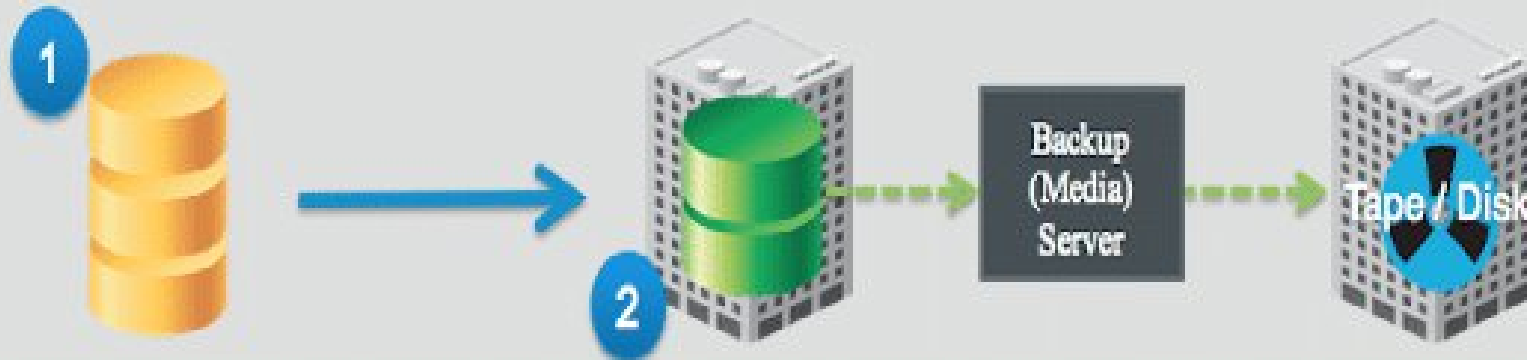
## Cloud Database

- Stored on **cloud** servers
- Easily scalable
- Pay-as-you-use model
- Automatic backup & recovery

## Traditional Databases

## Backup & Recovery Products

1990's to  
2000's



## Cloud Databases

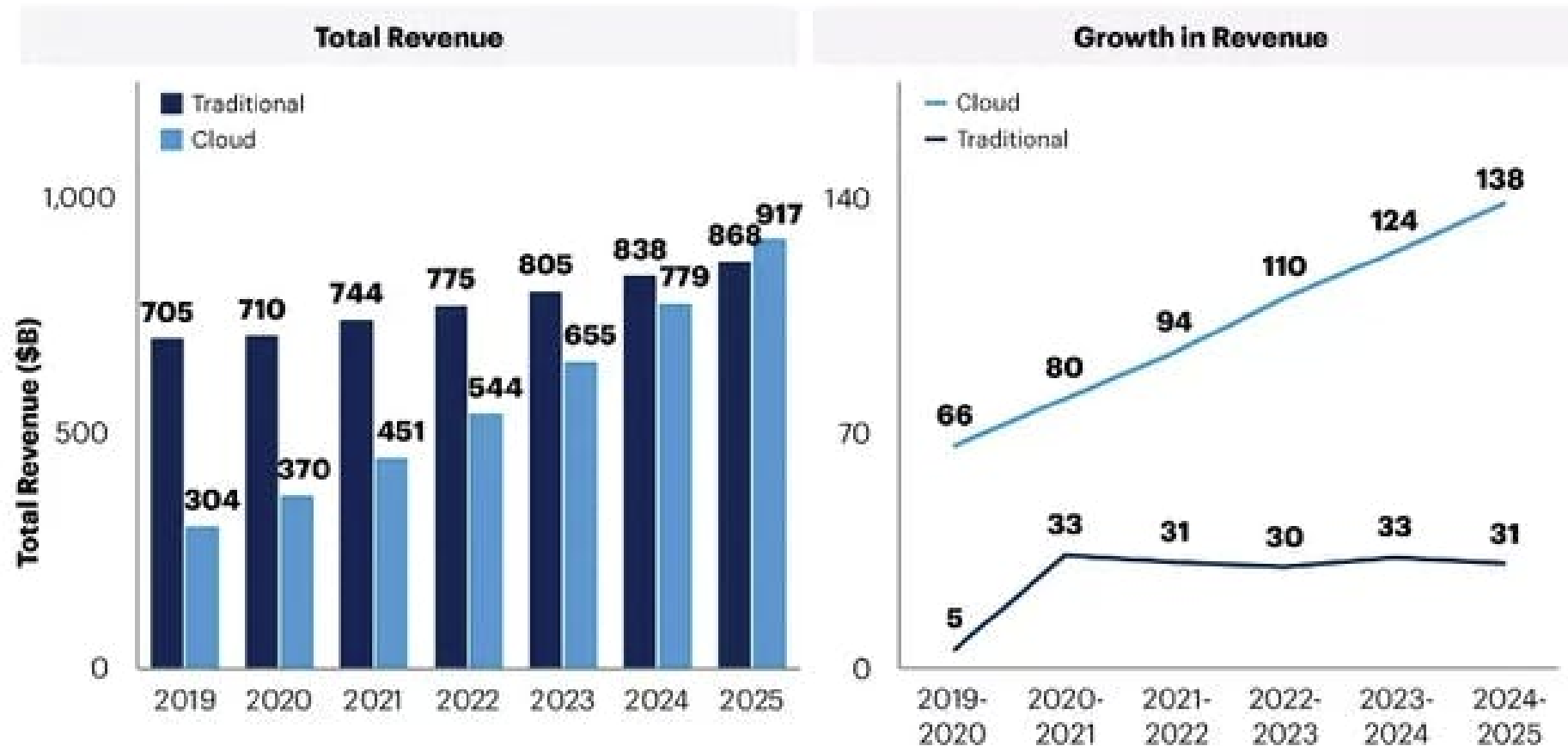
## Bring Forth The Needs To Innovate

2010's and  
Beyond



- 1 Scale-Out vs. Scale-Up
- 2 Local Storage vs. Shared Storage
- 3 Elastic vs. Static infrastructure

# Figure 1: Sizing Cloud Shift, Worldwide, 2019 – 2025



Source: Gartner  
759067\_C

Gartner.

## (a) SQL Cloud Databases (Relational)

- Uses tables, rows, and columns
- Follows ACID properties

### Examples:

- Amazon RDS (MySQL, PostgreSQL)
- Google Cloud SQL
- Azure SQL Database

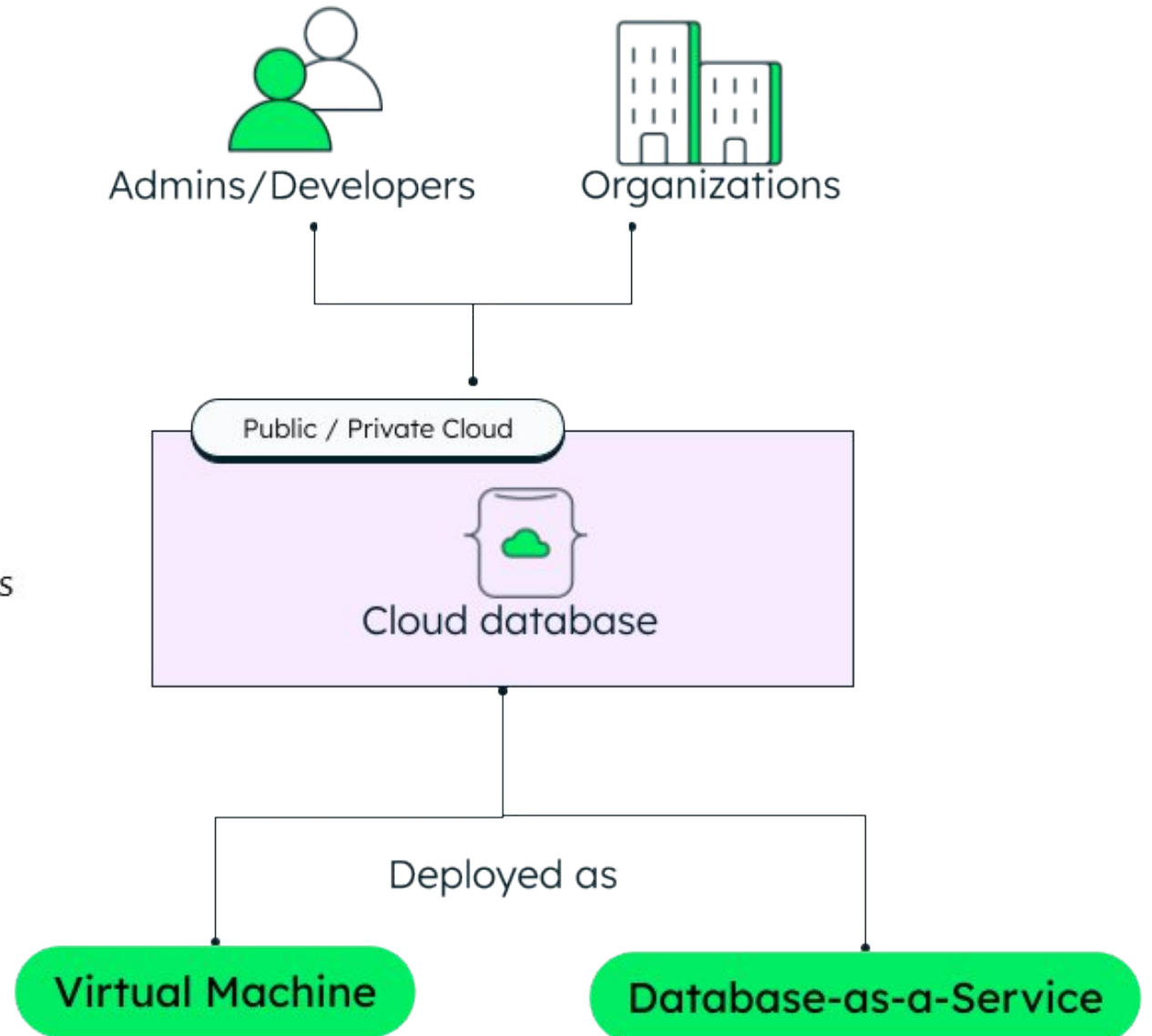
## (b) NoSQL Cloud Databases (Non-Relational)

- Uses documents, key-value, graph, or column models
- Suitable for big data & real-time apps

### Examples:

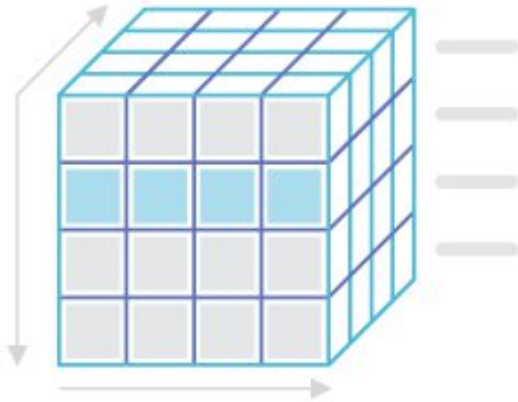
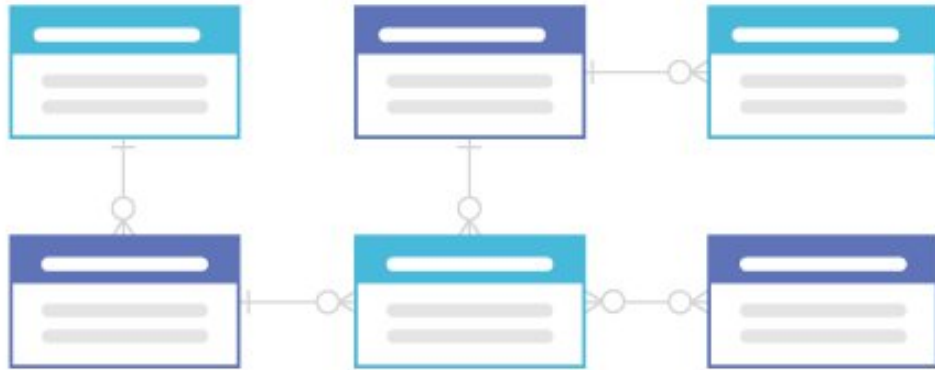
- MongoDB Atlas
- Amazon DynamoDB
- Firebase Firestore

# Types of Cloud Databases



## SQL

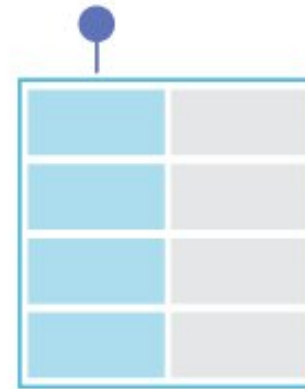
### *Relational Database Management Systems (RDBMS)*



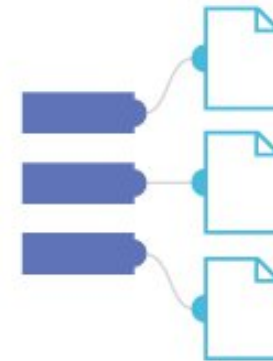
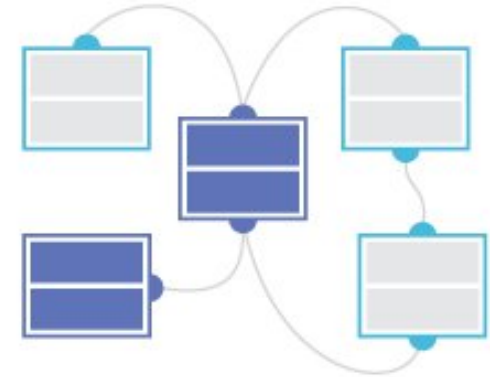
*Online Analytical Processing (OLAP) Cube*

## NoSQL

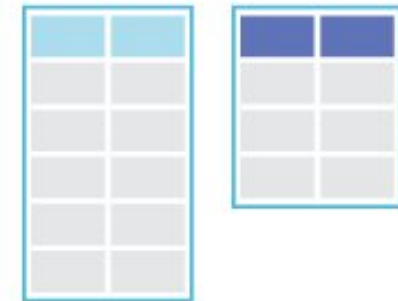
### *Key-Value*



### *Graph*



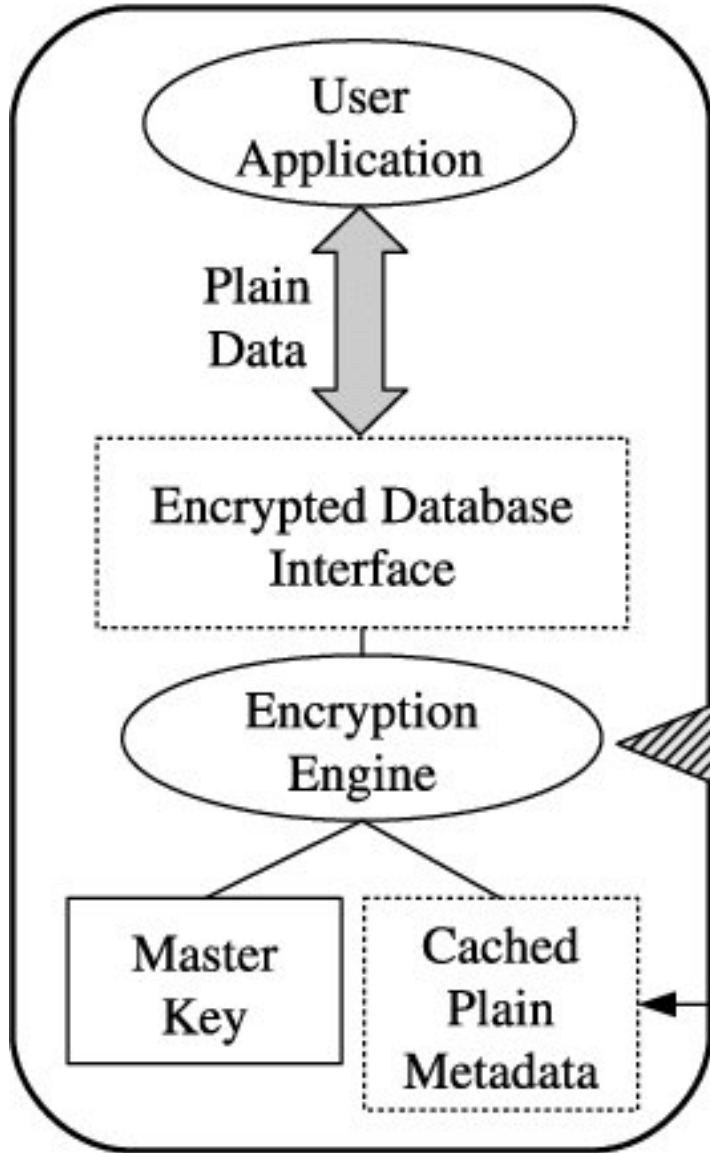
*Document*



*Column store*

# Architecture of Cloud Database

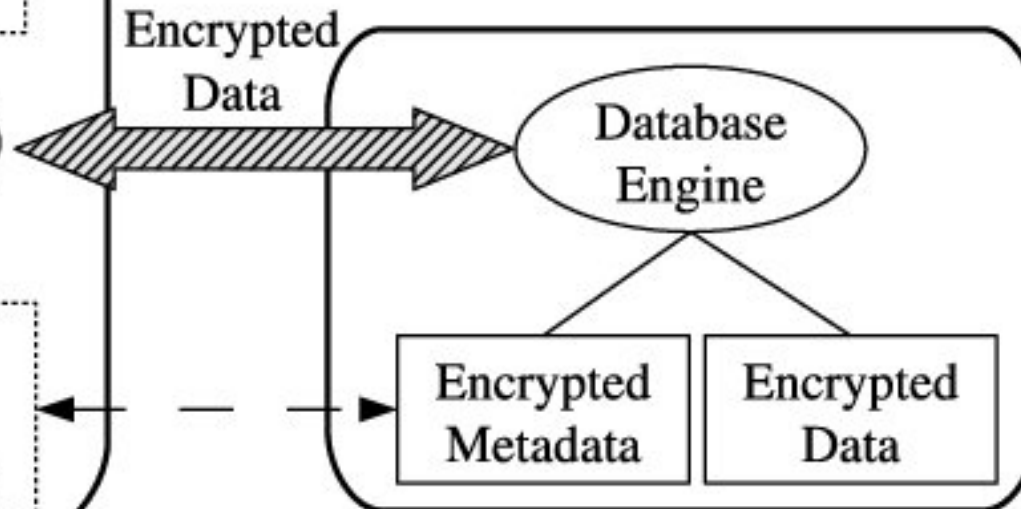
## Client



## How it works:

1. User sends request via browser or app
2. Request goes through the internet
3. Cloud provider processes the request
4. Data is fetched from cloud database
5. Result is returned to the user

## Cloud Database



## Examples of Cloud Databases

| Cloud Provider  | Cloud Database Example | Used For              |
|-----------------|------------------------|-----------------------|
| AWS             | Amazon RDS             | Web & enterprise apps |
| Google Cloud    | Cloud SQL              | Scalable applications |
| Microsoft Azure | Azure SQL Database     | Business systems      |
| MongoDB         | MongoDB Atlas          | Big data & analytics  |

### Advantages of Cloud Databases

- ✓ High availability
- ✓ Automatic backup & recovery
- ✓ Easy scalability
- ✓ Low initial cost
- ✓ Accessible from anywhere

### Disadvantages of Cloud Databases

- ✗ Internet dependency
- ✗ Security & privacy concerns
- ✗ Vendor lock-in
- ✗ Less control over infrastructure

## Examples of Cloud Databases

| Database Name    | Type (SQL/NoSQL)     | Managed By | Scalability | Max Database Size | Notable Users         |
|------------------|----------------------|------------|-------------|-------------------|-----------------------|
| Amazon RDS       | SQL                  | AWS        | Vertical    | 64 TB (Aurora)    | Netflix, Airbnb       |
| Amazon DynamoDB  | NoSQL (Key-Value)    | AWS        | Horizontal  | Unlimited         | Snapchat, Lyft        |
| Google Cloud SQL | SQL                  | Google     | Vertical    | 30 TB             | Spotify, HSBC         |
| Google Firestore | NoSQL (Document)     | Google     | Horizontal  | Unlimited         | Twitter, Alibaba      |
| Azure SQL DB     | SQL                  | Microsoft  | Vertical    | 4 TB (Hyperscale) | BMW, GE Healthcare    |
| Azure Cosmos DB  | NoSQL (Multi-Model)  | Microsoft  | Horizontal  | Unlimited         | Coca-Cola, ExxonMobil |
| MongoDB Atlas    | NoSQL (Document)     | MongoDB    | Horizontal  | Petabytes         | Adobe, Verizon        |
| Snowflake        | SQL (Data Warehouse) | Snowflake  | Horizontal  | Petabytes         | Capital One, DoorDash |

### Key Stats & Trends (2024)

- **Market Share:** AWS leads (~34%), followed by Azure (~22%) and Google Cloud (~11%).
- **Growth:** NoSQL databases (like DynamoDB & Cosmos DB) are growing at ~20% YoY.
- **Adoption:** Over **80% of enterprises** use at least one cloud database.

# Serverless Databases

- A serverless database is a cloud-managed database where the user does not need to manage servers, storage, scaling, or maintenance.
- The cloud provider automatically handles provisioning, scaling, backups, updates, and availability.

👉 *“Serverless” does NOT mean there is no server; it means you don’t manage the server.*

## How it Works

- You connect to the database using an API or endpoint.
- The database **automatically scales up or down** based on workload.
- You are charged **only for actual usage** (queries, storage, execution time).
- When idle, it can **pause or scale to zero** (in some services).

## Traditional vs Serverless Database Architecture

### Traditional DB

Fixed capacity

Manual maintenance

Always running

Pay for provisioned size

### Serverless DB

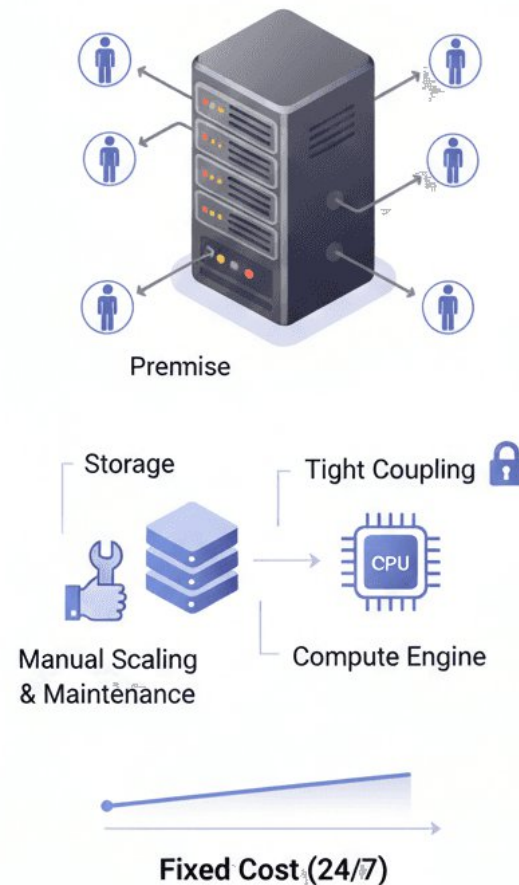
Auto-scaling

Fully managed

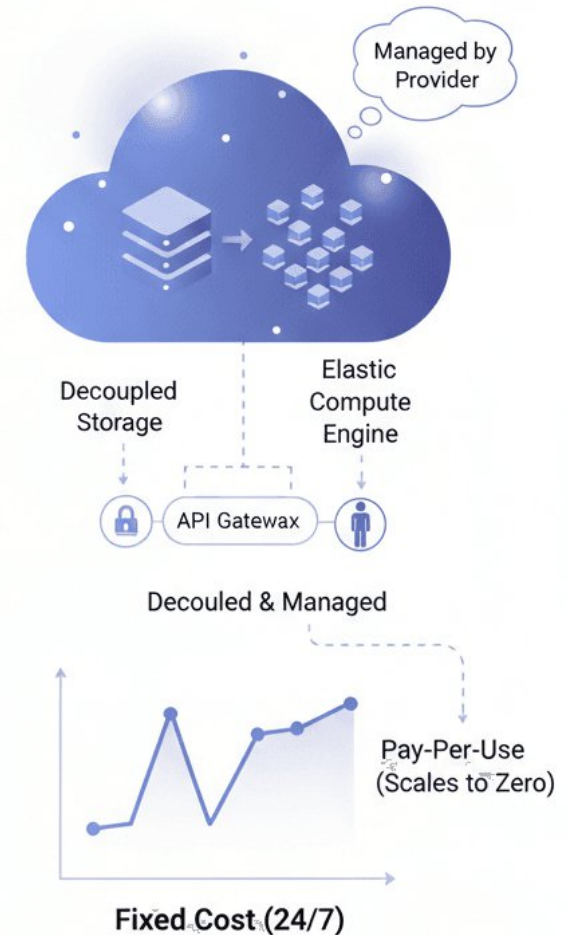
Runs on demand

Pay for actual use

### Traditional Database



### Serverless Database



## Popular Serverless Databases

- Amazon Aurora Serverless
- Google Firestore
- AWS DynamoDB
- Azure Cosmos DB
- FaunaDB
- Firebase Realtime Database

## Advantages

- ✓ No infrastructure management
- ✓ Cost-efficient for variable workloads
- ✓ Highly scalable
- ✓ Faster development
- ✓ Built-in backups & security

## Disadvantages

- ✗ Less control over tuning
- ✗ Cold-start latency (sometimes)
- ✗ Vendor lock-in
- ✗ Not ideal for constant heavy workloads

## Use Cases

- Web & mobile applications
- Startups & MVPs
- Event-driven systems
- IoT applications
- Serverless applications (Lambda, Cloud Functions)

## Simple Example

A college website with:

- Student registration
- Result lookup
- Online forms

*Traffic varies daily.*

*Serverless DB automatically scales during exam results day and scales down afterward, saving cost.*

# Vector Databases for AI/ML

- *A Vector Database is a specialized database designed to store, index, and search high-dimensional vectors (embeddings) efficiently.*
- *These vectors are generated by AI/ML models (NLP, CV, recommendation systems) and represent the semantic meaning of data.*

 *Instead of exact matching (SQL), vector databases perform similarity search.*

## Why Vector Databases are Needed in AI

Traditional databases:

- Work with **exact values** (=, <, >)

AI systems:

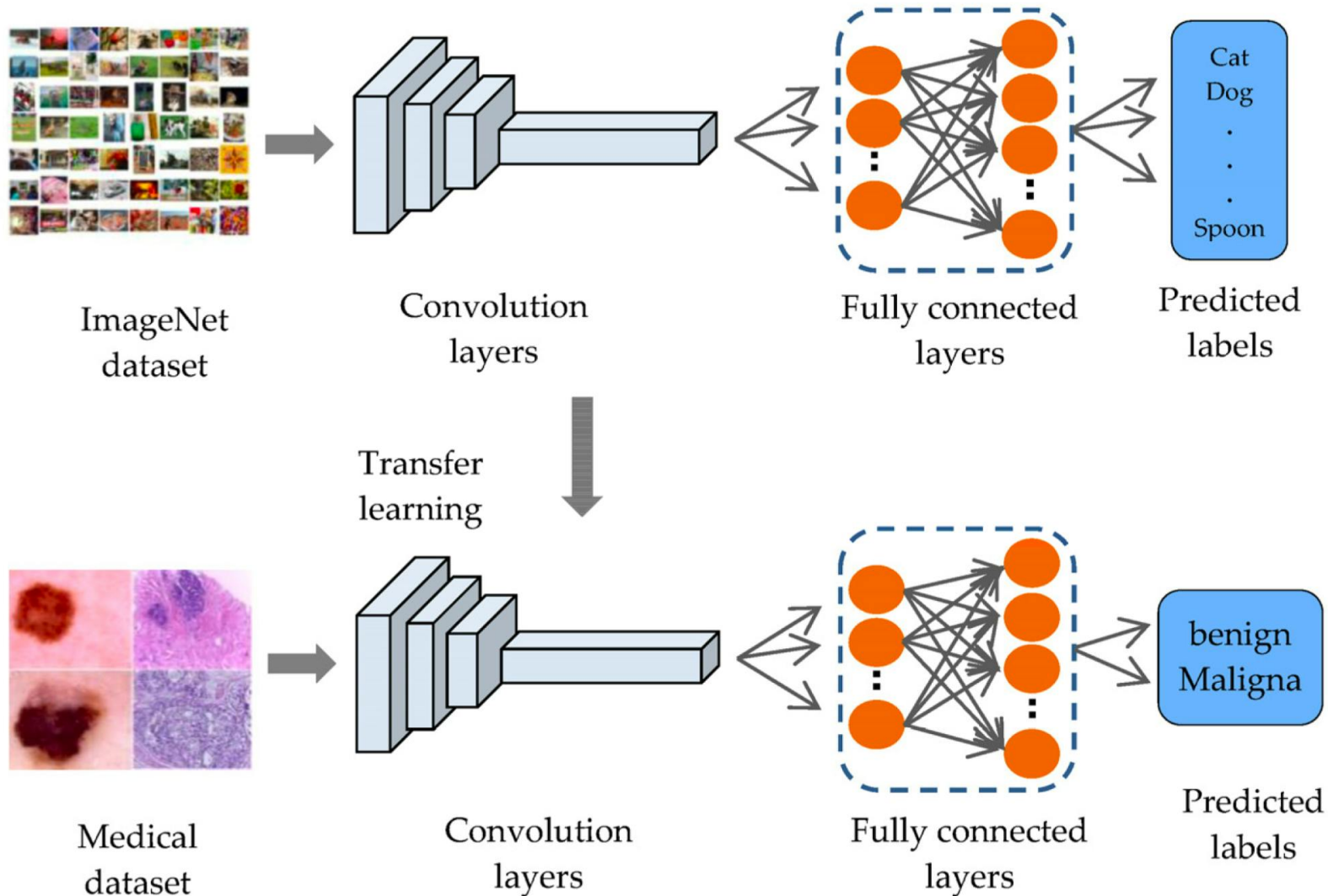
- Work with **meaning & similarity**

*Vector databases make this semantic comparison fast and scalable.*

Example:

- *"teacher"  $\approx$  "instructor"*
- *"car"  $\approx$  "vehicle"*

# How Vector Databases Work



## Step-by-step flow:

1. *Raw data (text, image, audio)*
2. *AI model converts data → vector (embedding)*
3. *Vectors stored in vector database*
4. *Query converted into a vector*
5. *Database finds nearest vectors*
6. *Returns most similar results*

*The model trained on ImageNet uses convolution layers to extract features, which are then applied to medical images to predict their class labels.*

## Key Features

| Feature            | Description                       |
|--------------------|-----------------------------------|
| Vector storage     | High-dimensional numeric arrays   |
| Similarity search  | Finds nearest vectors             |
| ANN indexing       | Fast Approximate Nearest Neighbor |
| Scalability        | Millions/billions of vectors      |
| Metadata filtering | Combine vectors + filters         |

## Similarity Measures

- **Cosine Similarity** – direction-based similarity
- **Euclidean Distance** – straight-line distance
- **Dot Product** – magnitude & direction

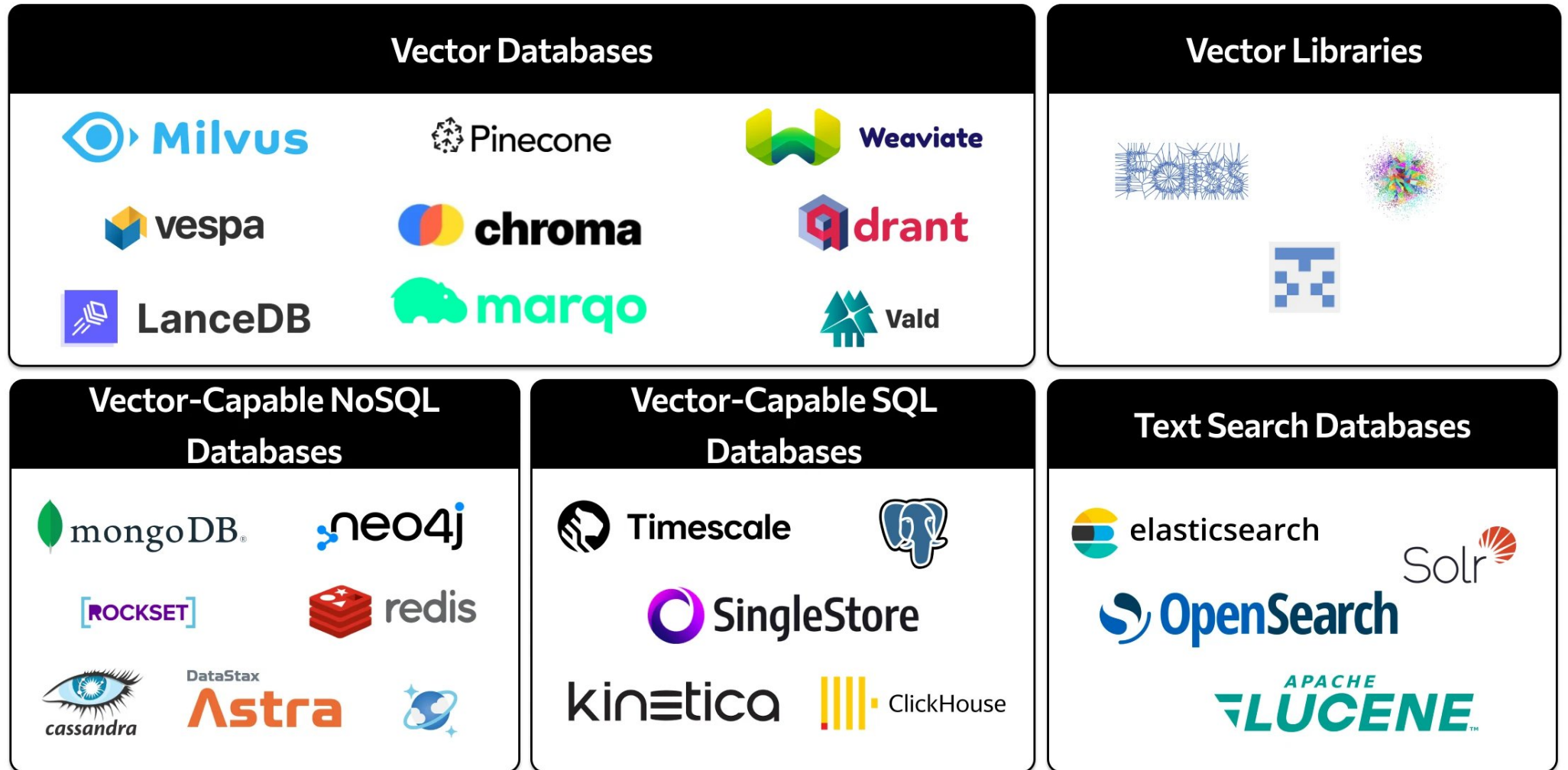
| Traditional DB  | Vector DB                 |
|-----------------|---------------------------|
| Exact match     | Semantic similarity       |
| Structured data | Unstructured data         |
| SQL queries     | Vector similarity queries |
| Tables          | Vector space              |

## Popular Vector Databases

- **Pinecone**
- **Milvus**
- **Weaviate**
- **FAISS (Meta)**
- **Chroma**
- **Qdrant**
- **ElasticSearch (Vector support)**

# Choosing a Vector Database

Top  
Vector  
Database  
for AI  
Apps



graft

## AI / ML Use Cases

### 1 Semantic Search

Search by meaning, not keywords.

- *"How to apply normalization in DBMS?"*
- Returns conceptually similar documents.

### 2 Recommendation Systems

- Product recommendation
- Video or course suggestions

### 3 Chatbots & RAG Systems

- ChatGPT-style apps
- Retrieval-Augmented Generation

### 4 Image & Face Search

- Find similar faces/images
- Reverse image search

### 5 Anomaly & Fraud Detection

- Detect unusual patterns
- Network security

Scenario:

**A college stores question answers & notes.**

Process:

- Notes → embeddings
- Stored in vector DB
- Student asks:  
*"Explain normalization in simple words"*

***System finds similar meaning answers, not exact keywords.***

## SQL vs Vector Query Example

SQL:

sql

```
SELECT * FROM notes WHERE topic = 'Normalization';
```

Vector Query (conceptual):

text

```
Find top 5 vectors closest to query vector
```

*A Vector Database is a specialized database that stores and searches high-dimensional vector embedding's using similarity measures, enabling semantic search and intelligent retrieval in AI/ML applications.*

## Advantages

- ✓ Understands meaning
  - ✓ Fast similarity search
  - ✓ Works with unstructured data
  - ✓ Essential for modern AI apps
- 

## Limitations

- ✗ Not for transactional data
- ✗ Requires embedding models
- ✗ Approximate results

# Edge Databases (IoT Applications)

*An Edge Database is a database that runs near the data source (edge devices) such as IoT sensors, gateways, cameras, or local servers, instead of a distant cloud.*

*🔑 Data is stored and processed locally at the edge, not always sent to the cloud.*

## Why Edge Databases are Needed in IoT

IoT devices generate **huge, continuous data**. Sending everything to the cloud causes:

- High **latency**
- High **bandwidth cost**
- Dependence on internet

Edge databases solve this by **processing data close to where it is generated**.

# Edge Computing Architecture

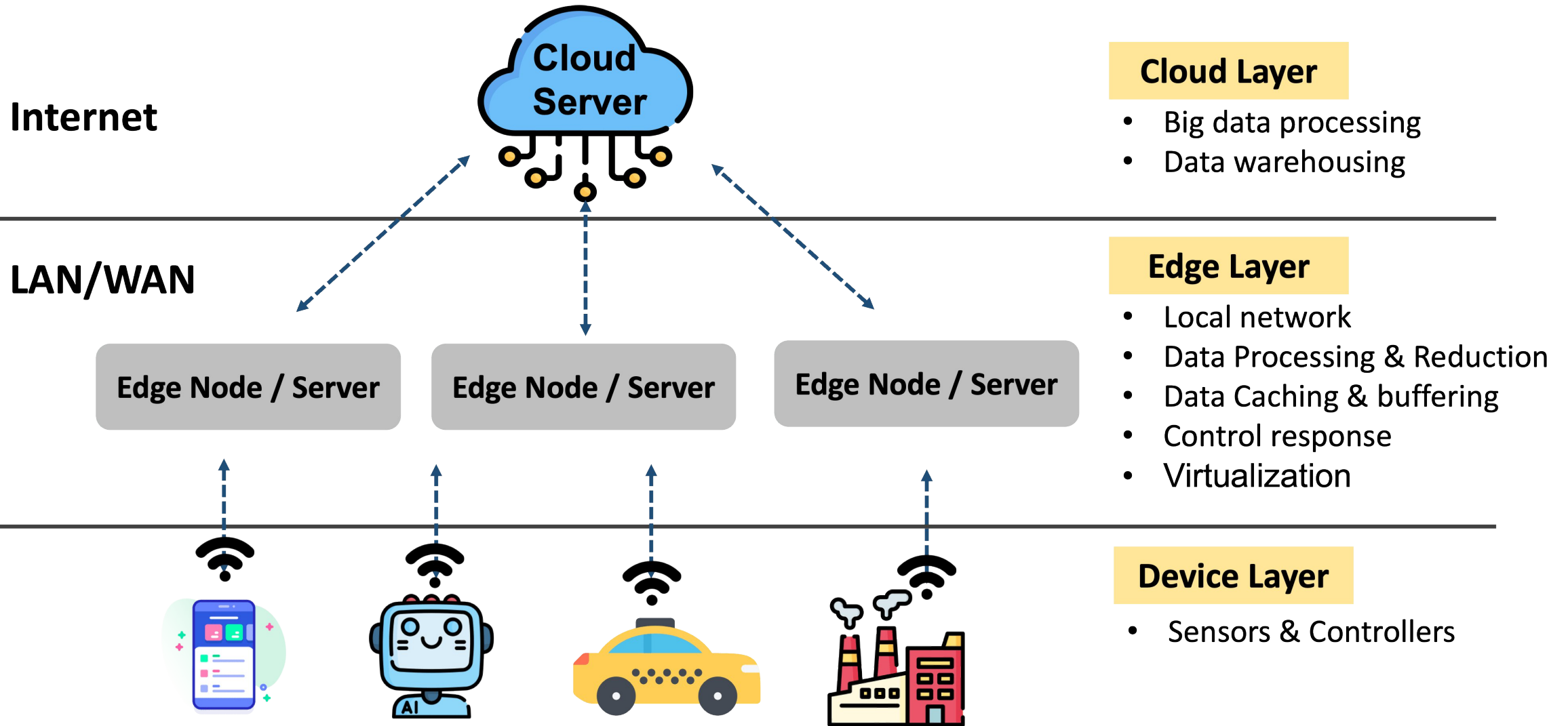
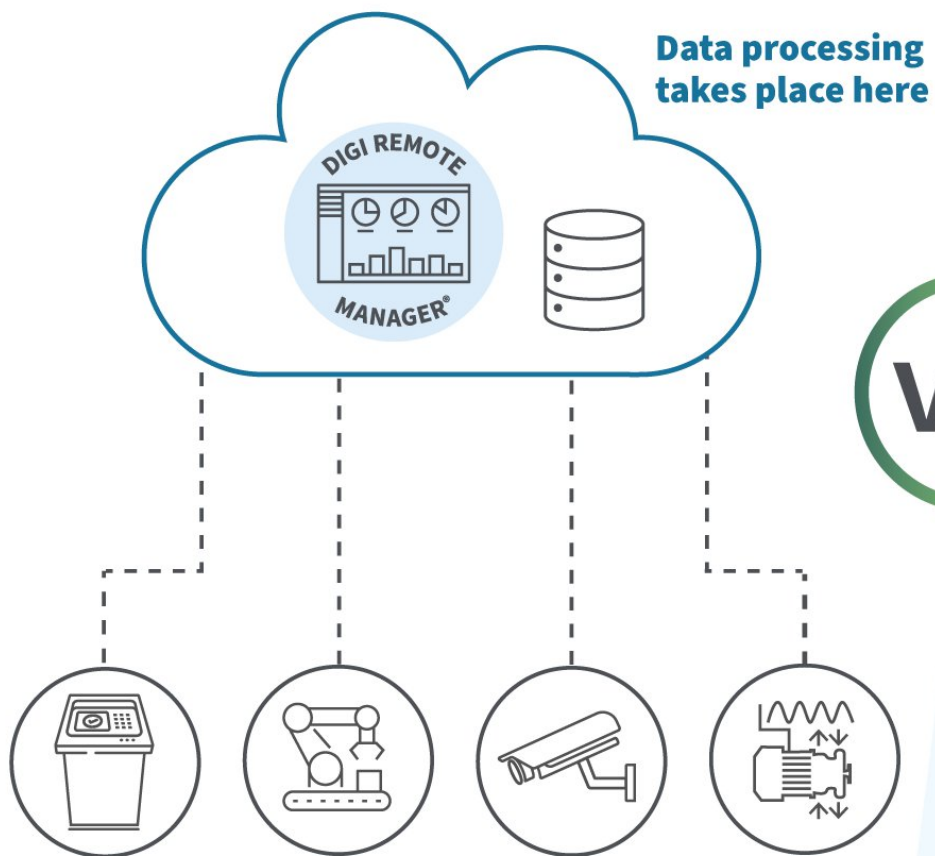


Figure : Edge computing architecture overview

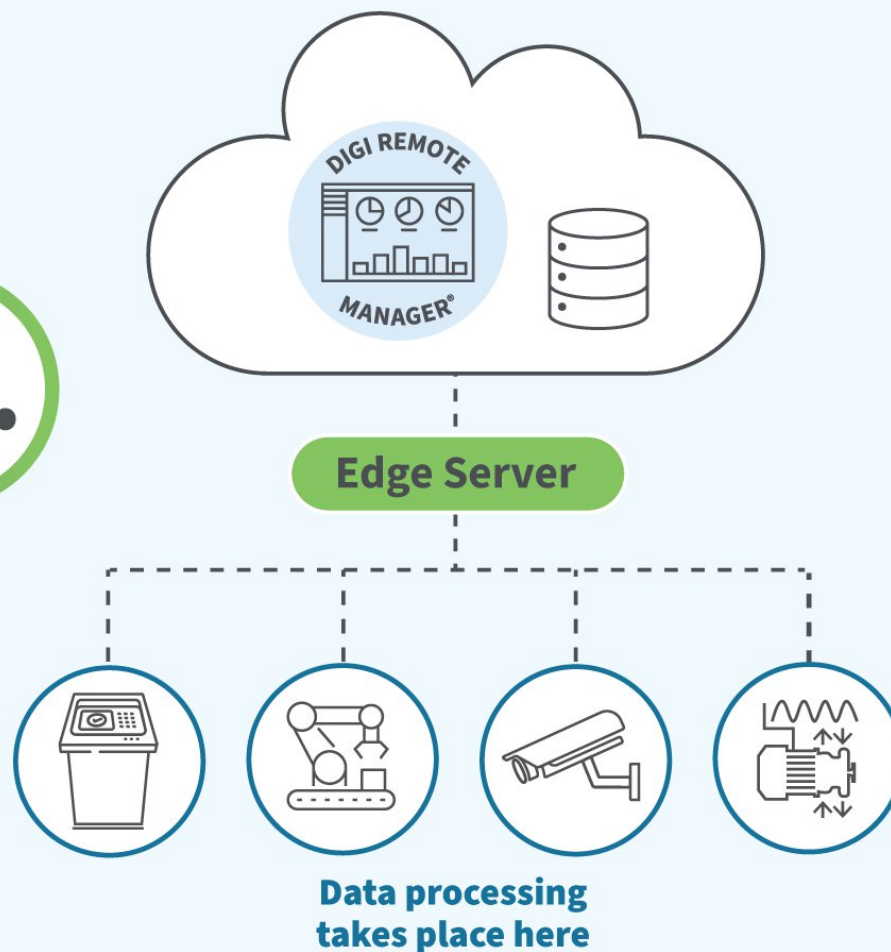
Source : The research team

# TRADITIONAL Cloud Computing



VS.

# MODERN Edge Computing



Edge Database

Runs near devices

Real-time processing

Works offline

Stores local data

Cloud Database

Runs in data centers

Higher latency

Needs internet

Stores global data

## How Edge Databases Work (Simple Flow)

1. Sensors generate data
2. Data is stored in an **edge database**
3. Immediate processing & decisions are made locally
4. Important/summary data is sent to cloud (optional)

## IoT Use Cases

### 1 Smart Healthcare

- Wearable sensors store patient data locally
- Immediate alerts for abnormal readings

### 2 Smart Factory

- Machines analyze sensor data locally
- Prevents failures in real-time

### 3 Smart Cities

- Traffic cameras process data locally
- Faster signal control

### 4 Agriculture (Farming)

- Soil & weather sensors store data locally
- Decisions without internet dependency

## Examples of Edge Databases

- SQLite
- InfluxDB Edge
- MongoDB Edge / Realm
- Redis Edge
- AWS IoT Greengrass (with local DB)

### *Simple Example*

*A temperature sensor in a cold storage:*

- *Stores readings in an edge database*
- *If temperature crosses limit → alarm triggered instantly*
- *Summary data sent to cloud daily*

## Advantages

- ✓ Fast response
- ✓ Low bandwidth usage
- ✓ Works without internet
- ✓ Better privacy

## Limitations

- ✗ Limited storage
- ✗ Limited computing power
- ✗ Data synchronization complexity

# Ethical Considerations in Database Engineering

*Ethical considerations in database engineering deal with the moral responsibilities of database designers, developers, and administrators while collecting, storing, processing, and sharing data. Since databases often store sensitive personal and organizational information, ethical handling of data is critical.*

## **Key Ethical Issues**

### **1. Privacy and Confidentiality**

Databases often store personal data like names, addresses, health records, and financial details.

Engineers must ensure this data is accessed only by authorized users.

*Example:* A hospital database should allow doctors to view patient records, but not allow non-medical staff to access medical history.

### **2. Data Security**

Ethical database design includes protecting data from unauthorized access, hacking, or data leaks using encryption, authentication, and access controls.

*Example:* Encrypting users' passwords in a banking database instead of storing them in plain text.

### **3. Data Accuracy and Integrity**

Engineers must ensure that stored data is correct, complete, and up-to-date, as wrong data can lead to serious consequences.

*Example:* Incorrect marks stored in a university database may affect a student's final result or career.

### **4. Access Control and Authorization**

Not all users should have the same level of access. Ethical practice demands role-based access.

*Example:* In a college database, students can view their own marks, teachers can update marks, but administrators manage the entire system.

### **5. Data Misuse and Bias**

Data should not be used for unethical purposes like discrimination, surveillance, or manipulation.

*Example:* Using customer data to unfairly deny loans based on caste, gender, or region is unethical.

### **6. Transparency and Consent**

Users should know what data is collected and how it is used. Their consent should be obtained.

*Example:* An e-commerce website informing users that their data will be used for order processing and recommendations.

**Example: Consider a student information system in a college:**

- *Student personal data is kept private (privacy).*
- *Passwords are encrypted (security).*
- *Marks and attendance are correctly updated (integrity).*
- *Different access levels for students, teachers, and admin (access control).*

## ***Real-World Ethical Dilemmas in Databases***

### **Final Thought:**

*"Just because you can store it,  
doesn't mean you should."*

– Ethical data engineers prioritize societal impact over unchecked data collection.

### **Case 1: Facebook-Cambridge Analytica Scandal**

- **Issue:** Unauthorized harvesting of 87M users' data for political profiling.
- **Ethical Violation:** Lack of transparency, misuse of personal data.

### **Case 2: Amazon's Biased Hiring Algorithm**

- **Issue:** AI recruitment tool discriminated against women.
- **Ethical Violation:** Training data reflected historical male dominance in tech.

### **Case 3: Clearview AI's Facial Recognition**

- **Issue:** Scraping billions of public photos without consent.
- **Ethical Violation:** Mass surveillance without user approval.

# Homework

17. Write short notes on: [PU, SOE]

- a. Vector Database
- b. Edge Database
- c. Cloud Database